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What is version control?

- ▶ Version control, also known as source control, is the practice of tracking and managing changes to software code.
- ▶ Version control software keeps track of every modification to the code in a special kind of database. If a mistake is made, developers can turn back the clock and compare earlier versions of the code to help fix the mistake.

Version Control Terminologies

- ▶ Repository
- ▶ Branch
- ▶ Local Repo
- ▶ Remote Repo
- ▶ Commit (snapshot or save records(changes))
- ▶ Clone (from local or remote)
- ▶ Push (upload local changes to remote)
- ▶ Pull (you pull changes from remote repo to your local)
- ▶ Pull request (tell other about your changes to pull it from local to remote)
- ▶ master / main

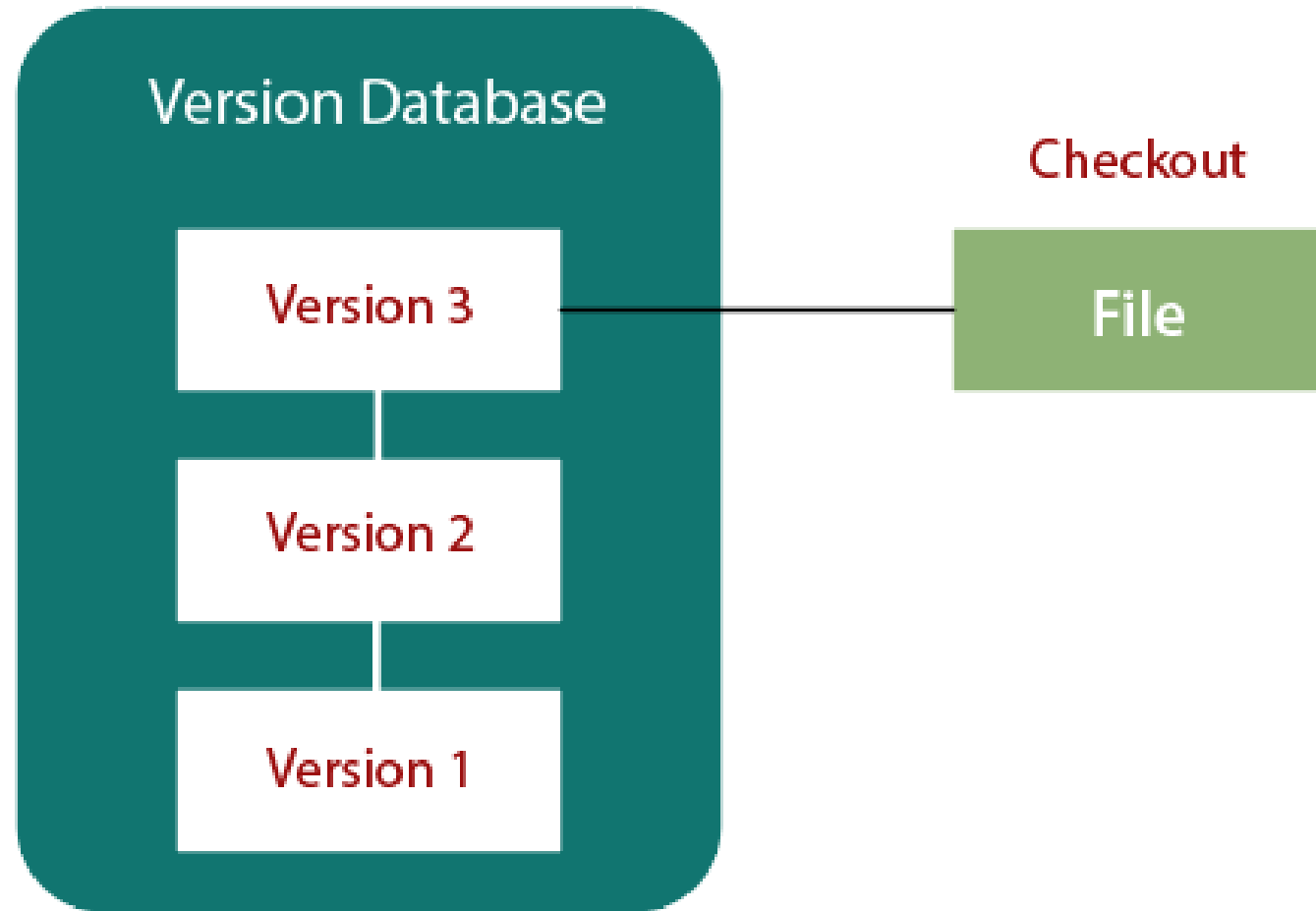


Types of Version Control Systems

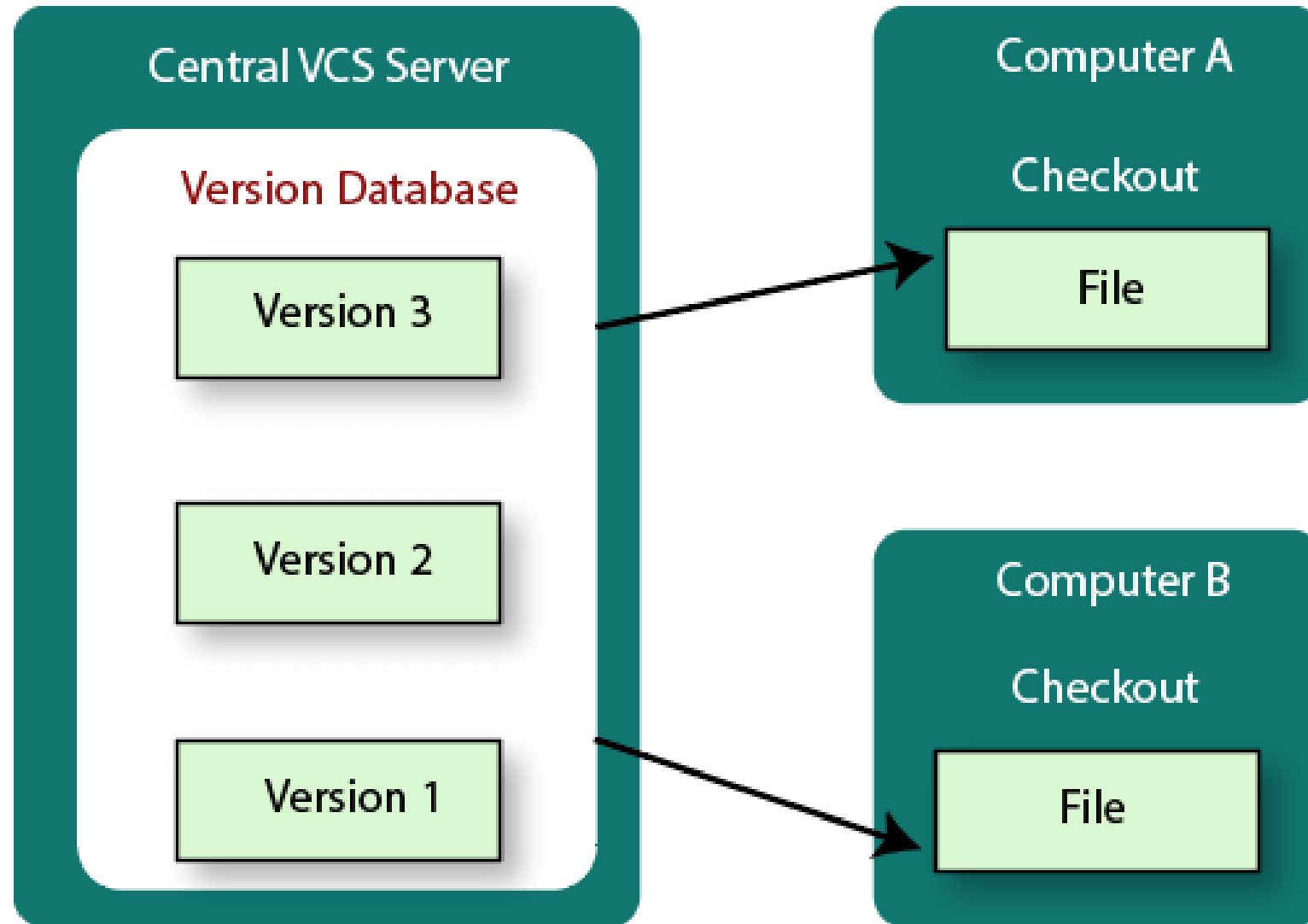
- 1- local version control
- 2- centralized version control
- 3- distributed version control

Local Version Control

Local Computer



Centralized version control systems are based on the idea that there is a single “central” copy of your project on a server, and programmers will “commit” their changes to this central copy.

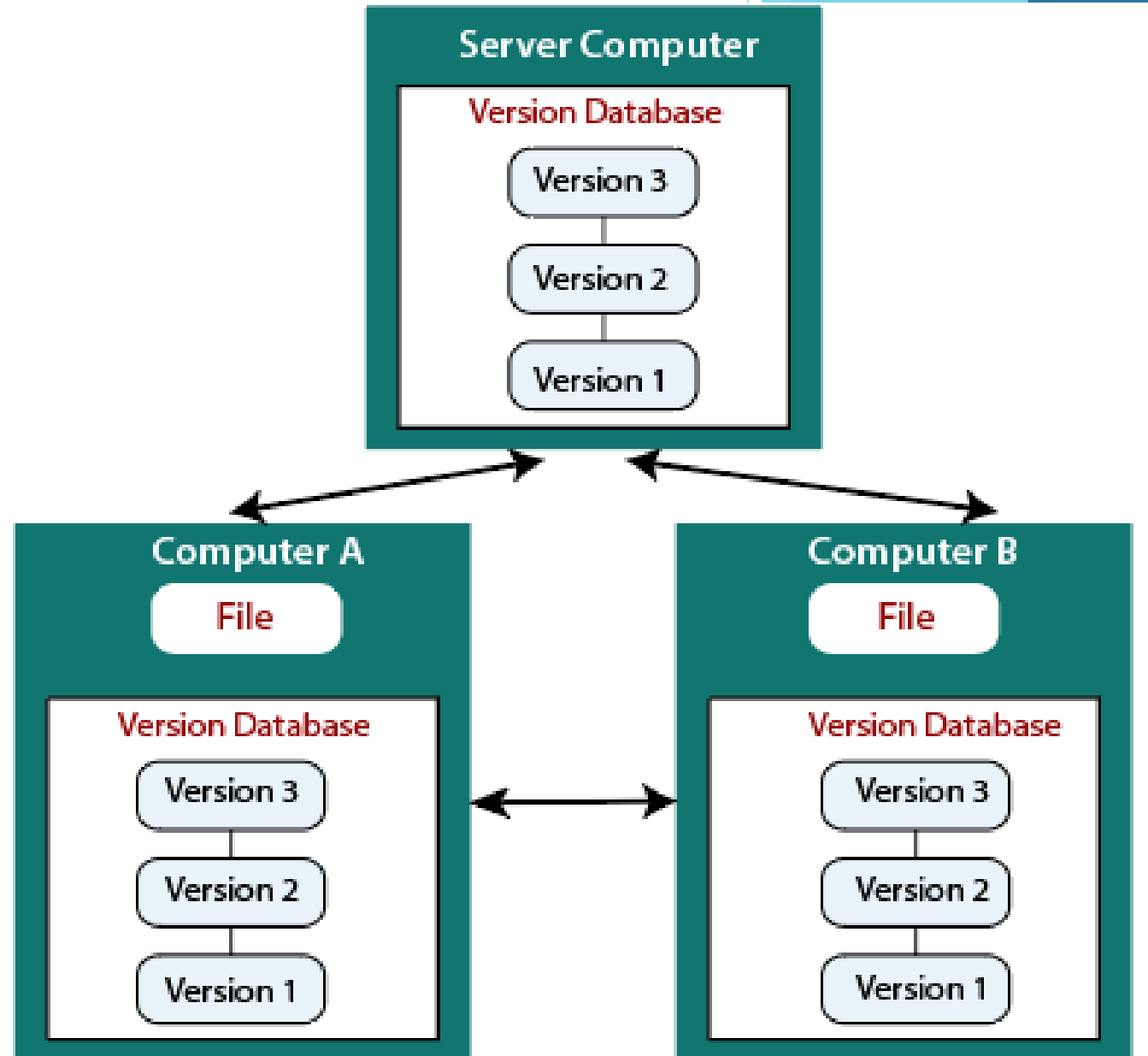


Disadvantages

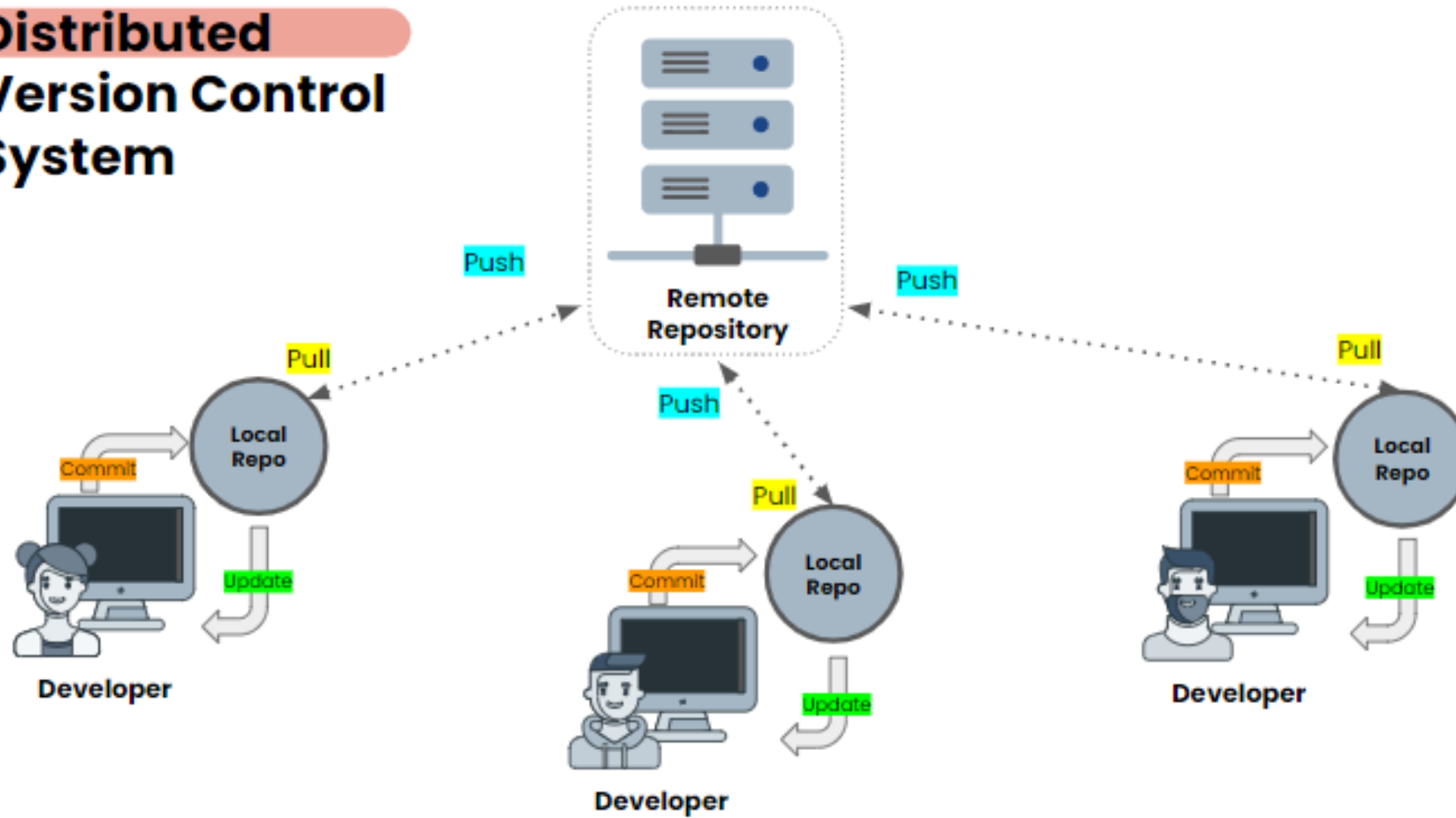
- ▶ If the main server goes down, developers can't save versioned changes.
- ▶ Need internet connection to commit the changes.

Distributed VCS

- ▶ These systems do not necessarily rely on a central server to store all the versions of a project's files. Instead, every developer “clones” a copy of a repository and has the full history of the project on their own hard drive.
- ▶ The act of getting new changes from a repository is usually called “pulling”, and the act of moving your own changes to a repository is called “pushing”.



Distributed Version Control System



Advantages

- ▶ Committing new changes can be done locally without anyone else seeing them. Once you have a group of changes ready, you can push all of them at once.
- ▶ Everything but pushing and pulling can be done without an internet connection. So you can work on a plane.
- ▶ Since each programmer has a full copy of the project repository, they can share changes with one or two other people at a time if they want to get some feedback before showing the changes to everyone.

Distributed VCS Examples

- ▶ Git



Git history



2002

Linux kernel project
began using
BitKeeper



2005

Linux kernel project
stopped using
BitKeeper

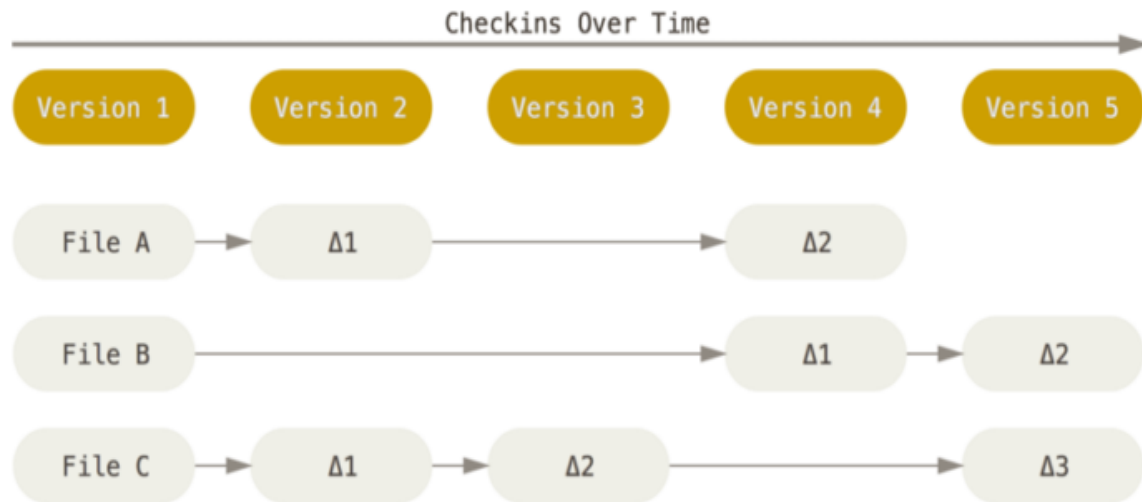


2005

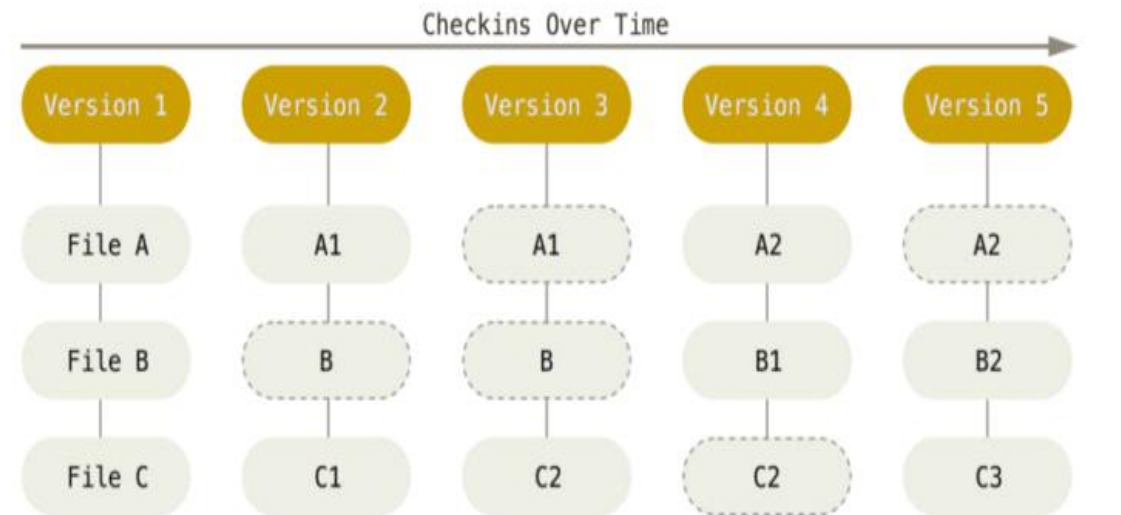
Linus Torvalds
started working on a
new DVCS called Git

Advantages of git

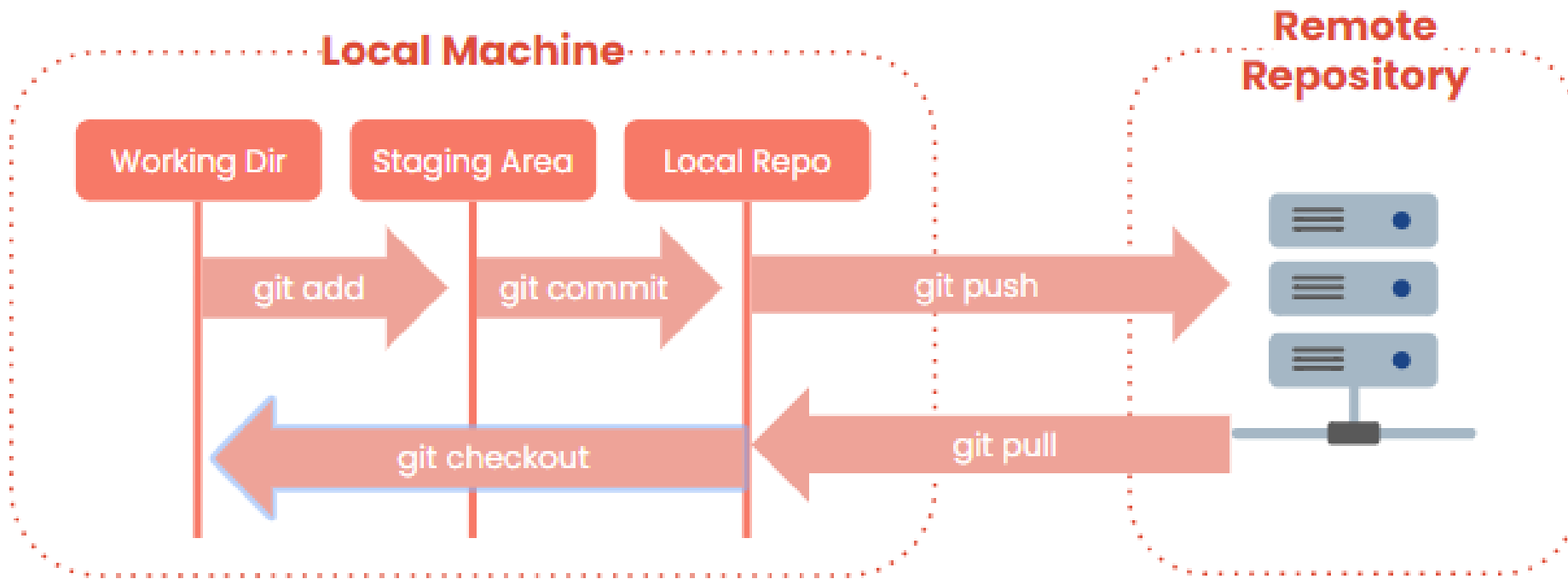
others VC



Git



- ▶ The Git Version Control System uses SHA-1 checksums on the contents of all change commits. In fact, the checksum is used as commit identifier and commonly referred to as "the SHA".
- ▶ Git's checksums include metadata about the commit including the author, date, and the previous commit's SHA.
- ▶ Git assures the integrity of the data being stored by using checksums as identifiers. If someone were to try to alter a commit or its meta data, it would change the SHA used to identify it. It would become a different commit.



Git file states

- ▶ Un Tracked (U)
- ▶ Git has three main states that your files can reside in: Tracked
- ▶ **Modified** means that you have changed the file but have not committed it to your database yet.
- ▶ **Committed (unmodified)** means that the data is safely stored in your local database.
- ▶ **Staged** means that you have marked a modified file in its current version to go into your next commit snapshot.

First Time Git Setup

- ▶ The first thing you should do when you install Git is to set your user name and email address.
- ▶ This is important because every Git commit uses this information, and it's immutably baked into commits you start creating.
- ▶ `git config --global user.name "your name"`
- ▶ `git config --global user.email "your e-mail"`

Starting a repo

- ▶ `mkdir firstdemo`
Make a directory
- ▶ `cd firstdemo`
Change directory to the above directory
- ▶ `git init`
Initialize an empty Git repository
- ▶ `ls -a`
List all the files & dir and the hidden files & dir

Create a new file

- ▶ `touch new.txt`
Create a new file called new.txt
- ▶ `echo "hello git" << new.txt`
write in new.txt file
- ▶ `git status`

```
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    new.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Add to Staging Area

- ▶ `git add new.txt` `/git add * /.`
 Staging the new.txt file
- ▶ `git status`

```
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   new.txt
```

Commit changes

- ▶ `git commit -m "the first commit"`
Commit the changes with msg
- ▶ `git status`

```
$ git status
On branch master
nothing to commit, working tree clean
```

Add & Commit

- ▶ `git commit -a -m "commit"`
- ▶ Add the changes & commit the changes in one line

- ▶ But , notice that this command doesn't add new files
- ▶ It only works with the changes that made inside the files itself

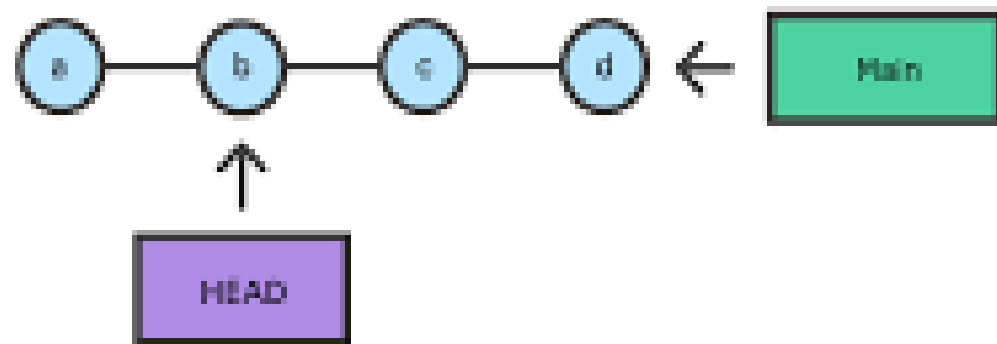
Git Logs

▶ git log

```
$ git log
commit bb2ffcbbc17d8eef90f64b9377c33e65536cda05 (HEAD -> master)
Author: nadamostafa <nadamostafa42127@gmail.com>
Date:   Wed Dec 7 01:55:54 2022 +0200

    initial commit
```

▶ git log --oneline



Git diff

- ▶ `git diff`

Show the unstaged differences since the last commit (working tree & stage area)

- ▶ `git diff --staged` `||` `--cached`

Show the staged differences since the last commit (stage area & repo)



Github

- ▶ is an Internet hosting service for software development and version control using Git
- ▶ Github is source for project and sources [gitlab, bitbucket]
- ▶ Github simplify using git



Clone a Remote Repo

- ▶ Https or ssh
- ▶ Git clone `https://github.com/****/*****`

To clone the entire repository to your local machine in a new directory

Git SSH Keys

- ▶ SSH keys come in pairs, public key that gets shared with services like GitHub and a private key that is stored only on your computer. If the keys match, you're granted access.
- ▶ Generate a new SSH key pairs
- ▶ `ssh-keygen -t ed25519 -C "your e-mail"`
- ▶ Copy the public key to your GitHub account
- ▶ `cat ~/.ssh/id_ed25519.pub`

Undoing things

- ▶ `rm -rf .git`
 - ▶ To remove local repo
- ▶ `Git restore new.txt`
 - To restore from stage area (ignore changes in working tree)
- ▶ `git commit --amend -m "your new msg"`
 - The above command will amend the added change to the last commit

Un staging changes

- ▶ `git restore --staged new.txt`
To unstage the changes
- ▶ `git reset HEAD script.py`
To unstage the changes

Why use version control?

- ▶ Developers contribute to the same project
- ▶ helps teams collaborate to fix issues & create new features
- ▶ You can control and backup your code (revert changes)
- ▶ Accelerates product delivery
- ▶ Keeping Track of All the Modifications Made to the Code
- ▶ Working on a new features without affecting the working code by branching

Lab

- ▶ Install Git.
- ▶ Create an account on GitHub.
- ▶ Create a new local repo and a remote repo on GitHub, then make a file contains your full name , then push it to the remote repo .
- ▶ Create a new remote repo on GitHub and clone it.